

Telugu Handwritten Character Segmentation Using YOLOv11



Guide: M Kiran Sastry
Branch of
Computer Science and Engineering

Team:
S Rohit Kumar - HU22CSEN0102032
P Sai Krishna - HU22CSEN0100302
K Mohan Krishna - HU22CSEN0100105
C Ganesh Sai Santosh - HU22CSEN0101777

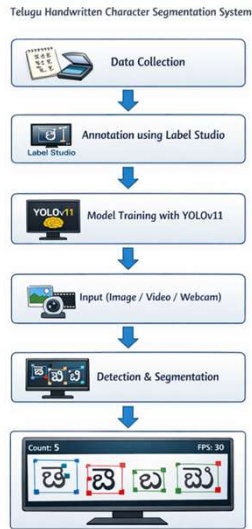
Abstract

Handwritten Telugu character segmentation is challenging due to complex shapes and overlapping characters. This work uses YOLOv11 to accurately detect and segment characters from handwritten text. A custom annotated dataset was used for training, and results show improved accuracy and efficient performance, making the approach suitable for document digitization.

Problem and Objective

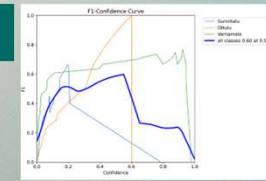
- Problem Statement:** Telugu handwritten text is difficult to segment due to curves and overlapping characters
- Existing methods show low accuracy
- Objective:** Build an accurate segmentation model using YOLOv11

Methodology



Implementation

- Image / Video / Webcam input
- Bounding box detection
- Confidence scores



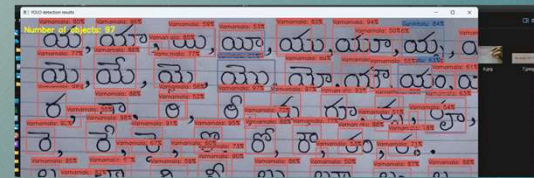
Results

Epoch: 3/100 GPU mem: 7.14G box_loss: 1.253 cls_loss: 1.432 dfl_loss: 1.477 Instances: 4533 size: 640x640

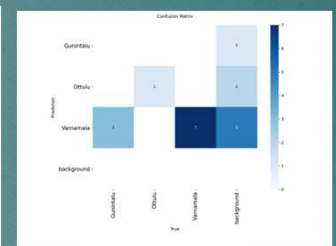
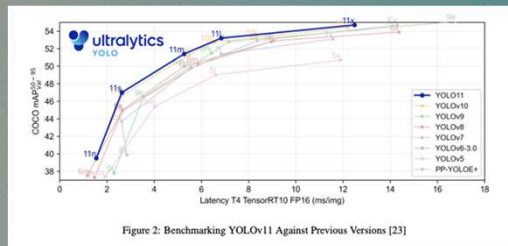
Epoch	GPU mem	box_loss	cls_loss	dfl_loss	Instances	size
3/100	7.14G	1.253	1.432	1.477	4533	640x640
Class	Images	Instances	BoxCP			
all	2	11	0.00225	0.159	0.00027	0.00218

Epoch: 3/100 GPU mem: 7.14G box_loss: 1.253 cls_loss: 1.432 dfl_loss: 1.477 Instances: 4533 size: 640x640

Epoch	GPU mem	box_loss	cls_loss	dfl_loss	Instances	size
3/100	7.14G	1.253	1.432	1.477	4533	640x640
Class	Images	Instances	BoxCP			
all	2	11	0.00225	0.159	0.00027	0.00218



Comparison Graph Of YOLO Models



Dataset Preparation

- 1.Data collection
- 2.Annotation using Label Studio



References

- [1] Ultralytics, "YOLO Documentation," Ultralytics, 2024. [Online]. Available: <https://docs.ultralytics.com/>
- [2] Label Studio, "Label Studio Documentation," Heartex, 2024. [Online]. Available: <https://labelstud.io/>
- [3] Anaconda, "Anaconda Documentation," Anaconda Inc., 2024. [Online]. Available: <https://docs.anaconda.com/>
- [4] Website: <http://www.youtube.com/@EdgeElectronics>
- [5] https://youtu.be/rORspILG260?si=_V09snYqJMOUdpog
- [6] G. Bradski, "The OpenCV Library," Dr. Dobb's Journal of Software Tools, 2000.
- [7] S. van der Walt, S. C. Colbert, and G. Varoquaux, "The NumPy Array: A Structure for Efficient Numerical Computation," Computing in Science & Engineering, vol. 13, no. 2, pp. 22-30, 2011.

Future Scope

- 1.Extend the system to full word and sentence recognition (complete OCR pipeline)
- 2.Train on a larger and more diverse dataset with real-world handwritten samples
- 3.Improve accuracy using advanced versions of YOLO or hybrid models
- 4.Implement character recognition (classification) after segmentation
- 5.Develop a mobile or web-based application for real-time usage
- 6.Integrate with document digitization systems for practical deployment

Conclusion

This project demonstrates an effective approach for Telugu handwritten character segmentation using YOLOv11. By leveraging a properly annotated dataset and a deep learning-based detection model, the system achieves improved accuracy and reliable real-time performance. Compared to earlier attempts, the proposed method provides better handling of handwriting variations and complex character structures, making it suitable for practical applications such as document digitization and handwritten text analysis.